

MUKAM Augusta Priscille

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 **Bordeaux, France**
Research Engineer / Postdoc
— open to relocation and
international opportunities

Research Engineer – Distributed Systems & Cloud | PhD (University of Bordeaux, 2026)

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 research.amukam.com/

Main Skills

- Operating systems**
Linux: Ubuntu, Kali Linux, Windows
- Programming**
Rust, R, Go, C, Python, C++, Java, Js, HTML, CSS, Django, MATLAB
- Editing**
Latex, Microsoft office tools, Google suite
- Cloud & Systems Infrastructure**
Grid5000, Ansible, Docker, Kubernetes, Terraform
MySQL, MariaDB, MongoDB, Cassandra, Kafka
AWS, GCP, Azure
- Modeling and Management**
DevOps, Agile, Scrum, Jira, Slack
- Soft Skills**
Cross-disciplinary technical profile · Scientific writing & communication · Analysis & synthesis
Communication, Proactivity, Discipline, Adaptability, Focus.

Education

2022–2026: University of Bordeaux, France

PhD in Distributed Systems —
Thesis: “Byzantine-Resilient Peer Sampling for Large-Scale Distributed Systems”

Summary

PhD in Distributed Systems (University of Bordeaux, April 2026). Research expertise in Byzantine fault-tolerant gossip protocols, scalable peer sampling, and memory-efficient distributed algorithms for blockchain and large-scale P2P systems. Strong practical background in systems programming (Rust, Go, C), cloud infrastructure (AWS, GCP, Azure, Kubernetes, Docker), and large-scale distributed experimentation (Grid5000, Ansible). Targeting Research Engineer or Postdoc positions in distributed systems, cloud computing, or adjacent fields (ML infrastructure, networked systems, applied security).

Work Experience

Fellow Teaching and Research Associate University of Bordeaux, France	01/26 - 08/26
<i>168 hours</i> L3 : Compilation, L2 : Algorithms for Tree Data Structures, M1 : Programming Project	
PhD researcher LaBRI, University of Bordeaux, France	10/22 - 04/26
Designed and evaluated Byzantine fault-tolerant gossip protocols for peer sampling in large-scale distributed systems and blockchain networks. Key results: up to 60% improvement in fault tolerance (robust with 40% adversarial nodes); scalable hash-based frequency estimation with minimal memory overhead. <i>Supervisors:</i> Laurent Réveillère , Joachim Bruneau-Queyreix .	
R&D Intern Digital Power team, Schneider Electric, Grenoble, France	02/22 - 08/22
- Secure storage in Am65xx devices - TrustZone with OpTEE, Linux filesystem-based Secure Storage - Secure application written for SHA256, AES 128 CBC, RSA, ECDSA calculations and TLS connections based on OpenSSL and Trusted Platform Module (TPM) . <i>Supervisors:</i> Julien Michel , Milan Stevanovic <i>Achievements:</i> Enabled Secure boot on Am65xx HS Schneider devices	
Research Intern LIP laboratory, ENS Lyon, France	03/21 - 09/21

funded by the ANR GenBlock project

Defended: **April 2026**

Gossip-based peer sampling · Byzantine fault tolerance · Scalable frequency estimation · Blockchain security

2021-2022: Grenoble Inp - Ensimag - France

Master of Science in Informatics at Grenoble (MoSiG) in Cloud and Distributed Systems

LabEx MILYON Scholarship of excellence

Large scale Data-Management and Distributed systems, Cloud computing, Advanced parallel system, Advanced networking

2016-2021: National Advanced School of Engineering of Yaounde (ENSPY) - Cameroun

Master of Engineering in Computer Science

Software engineering and programming, Algorithms, security, Operation research and graph, Machine Learning and Data Mining, Management

Research Interests

Distributed systems · Byzantine fault tolerance · Cloud infrastructure · Systems programming · ML infrastructure · Applied security · Networked systems

Languages

French	Fluent
English	Fluent
Spanish	Beginner

Hobbies

Fitness, Singing, Travelling, Reading, Learning

- Study of Intel SubPage Write Protection (SPP) Intel specification
 - Implemented SPP handling functions in the XEN hypervisor
 - Bytecode analyzer based on the LLVM compiler
 - A study of vulnerabilities over the last 12 years.
- Supervisors: [Alain Tchana](#), [Stella Bitchebe](#)
- Achievements: **SPP-based memory protection policies against buffer overflows**

Publications

- [1] A. Mukam et al., “Robust frequency estimation for peer sampling in adversarial context,” 2026, under publication
- [2] A. Mukam et al., “AUPE: Collaborative Byzantine Fault-Tolerant Peer-Sampling,” NCA 2024 – 22nd IEEE Int. Symp. on Network Computing and Applications, Bertinoro, Italy. Acceptance rate: 33% · 🏆 Best Paper Award

Certifications

Kubernetes Fundamentals (LFS258) for Certified Kubernetes Administrator (CKA)	The Linux Foundation (ongoing)
Introduction to Cloud Infrastructure Technologies LFS151	The Linux Foundation
DevOps Basics for Everyone IBM CD0115EN	edX
Introduction à la programmation orientée objet (en Java) Score : 93.40%	Coursera

Teaching & Service

Interviewer for the ETE Community Experience Feedback	2026
Session technician at Compas'25	2025
Session chair at Wise'25	2025
Bordeaux INP - Enseirb Matmeca, Telecom, Bordeaux, France	2023-2025
Teaching assistant (50 hours*2) and Internship tutoring (M1 and M2 internships) L3: Imperative Programming, Introduction to Operating Systems M1: Network Programming	
L1 Student mentoring on Large scale experiments in G5K	2023
Popularising research to high school and License students	2023-2024